

Exercise Sheet 2

Linear Regression

task 1 Linear regression

- (a) Given are measured values (x_i, y_i) , $i = 1, \dots, m$. We are looking for the equalization line $g(x) = ax + b$ with parameter set (a, b) , so that the error function

$$f(x) = \sum_{i=1}^m (ax_i + b - y_i)^2$$

becomes minimal. Show that for the linear system of equations

$$\begin{pmatrix} \sum_{i=1}^m x_i^2 & \sum_{i=1}^m x_i \\ \sum_{i=1}^m x_i & m \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^m x_i y_i \\ \sum_{i=1}^m y_i \end{pmatrix}$$

must be solved.

- (b) Write a method that calculates an fitting line for m tuples of measured values.
- (c) Test your method for a data set with $m = 100$. Select your data set randomly (`np.random.random`) and sort both the x and the y components so that $x_i \leq x_j, i < j$ and $y_i \leq y_j, i < j$ applies. Plot the data points and the equalization line graphically using `matplotlib`.
- (d) Now the straight line should also pass through the point (s, z) . How does this change the system to be solved? Is there always a solution? Graph your results here too and compare them for the same data set.

Imagine you work in a winery: At your last wine tasting, you noted the quality of the wines ($m = 1507$) in a table (`winedata.csv`) and collected various parameters (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulphur dioxide, total sulphur dioxide, density, pH, sulphates, alcohol) for the properties of the respective wine. They now want to determine a correlation between the various properties and the quality using a training data set. The aim is to be able to estimate the quality of future wines as part of the test data set. To do this, use a linear regression in several dimensions. Use the template `Exercise02Template.ipynb`.

- (a) Load the data using the `pandas` library. The first 11 columns are the parameters, in the 12th column you will find the quality.
- (b) Fit the parameters $a \in \mathbb{R}^{11}$ and $b \in \mathbb{R}$ so that for the training data the optimization problem

$$\|X \cdot a + b - Y\|^2 \rightarrow \min$$

is solved. Use the information from task 1 a) and set up an LGS first.

- (c) Validate your prediction $X \cdot a + b$ with the `LinearRegression` module from the library `sklearn.linear_model`.
- (d) Test your regression parameters using the test set. How good is your prediction?